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Department of Biology

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MAJOR PROFESSIONAL INTERESTS

Evolutionary genetics; quantitative genetics; molecular genetics; parasitology; drug resistance

EDUCATION

2008-2013 Post-doctoral fellowship
Princeton University
Advisor: Dr. Leonid Kruglyak

2000-2008 Ph.D. in Biology
Massachusetts Institute of Technology (MIT), Cambridge, MA
Advisor: Dr. H. Robert Horvitz
Dissertation: The synthetic Multivulva genes and their suppressors regulate opposing cell fates through chromatin remodeling

1996-2000 B.S. in Biological Sciences with departmental honors
Stanford University, Stanford, CA
Advisor: Dr. Matthew P. Scott
Dissertation: *in vivo* analysis of *Drosophila* heart-tube formation

AWARDS, HONORS, AND FELLOWSHIPS

2026 Johns Hopkins Discovery Award recipient
2022 - 2024 Fulbright Global Scholar (Chile, New Zealand, Taiwan), declined
2022 - 2024 Editorial Board Member of *Journal of Nematology*
2021 Distinguished Teaching Award, Weinberg College of Arts and Sciences
2020 - 2023 Editorial Board Member of *Gene*
2019 - 2022 Human Frontiers Science Program Award Recipient
2018 - 2023 National Science Foundation CAREER Award Recipient
2017 - 2020 Editor Board Member of *BMC Genomics*
2015 - 2019 American Cancer Society Research Scholar
2015 - 2017 March of Dimes Basil O'Connor Research Scholar
2015 - Editorial Board Member of *Trends in Genetics*
2014 - 2018 Pew Scholar in the Biomedical Sciences
2012 - 2013 Howard Hughes Medical Institute Post-doctoral Fellow
2011 - 2012 National Cancer Institute Post-doctoral Fellow, training grant T32-CA009528
2009 - 2011 Ruth L. Kirschstein National Research Service Award Recipient
2005 - 2006 Anna Fuller Cancer Graduate Research Fellowship
2000 Firestone Medal for Excellence in Undergrad. Research (top Biological Sciences researcher)
1999, 1998 Howard Hughes Medical Institute Summer Research Fellowship
1998 Stanford University Undergraduate research small grant recipient
1996-1998 Robert C. Byrd Honors Scholarship recipient

EMPLOYMENT

2025 - Vice Chair, Department of Biology, Johns Hopkins University
2023 - Professor of Biology, Johns Hopkins University
2022 - 2023 Professor of Molecular Biosciences, Northwestern University
2022 - 2023 Professor of Cell and Developmental Biology, Feinberg School of Medicine, Northwestern University
2020 - 2022 Associate Professor of Cell and Developmental Biology, Feinberg School of Medicine, Northwestern University
2019 - 2022 Associate Professor of Molecular Biosciences, Northwestern University
2013 - 2023 Preceptor for the Interdisciplinary Biological Sciences Graduate Program (IBiS)
Resident member of the Chemistry of Life Processes Institute (CLP)
Full Member of the Robert H. Lurie Comprehensive Cancer Center
Member of Northwestern Institute on Complex Systems (NICO)
2013 - 2019 Assistant Professor of Molecular Biosciences, Northwestern University
2008 - 2013 Post-doctoral fellow, Princeton University, Advisor: Dr. Leonid Kruglyak
2000 - 2008 Graduate student, Biology Department at Massachusetts Institute of Technology (MIT), Advisor: Dr. H. Robert Horvitz

RESEARCH SUPPORT

PRESENT

2026 - 2027 Johns Hopkins University Discovery Award
Enabling parasitic roundworm genetics and genomics for discovery of drug resistance loci
Lead PI: Andersen, co-PIs: Timp, Carlton

2026 - 2029 National Science Foundation, Capacity DBI (2534139)
Critical Expansions and Improvements to the Caenorhabditis Natural Strain Resource
PI: Andersen

2020 - 2031 National Institutes of Health (R01 AI153088) - NIAID
Discovery of novel benzimidazole resistance mechanisms
Lead PI: Andersen, co-PIs: Fraser (UCSF), Schroeder (Cornell) Zamanian (UW Madison)

2025 - 2028 Boehringer Ingelheim Research Support
Heterakis spumosa, a novel ascarid parasite model
PI: Andersen

2025 - 2028 Gordon and Betty Moore Foundation Award
PI: Andersen

2025 - 2028 National Science Foundation, EDGE CMT (2524223)
Defining the mechanistic roles of hyper-divergent regions in gene-by-environment interactions
PI: Andersen

2025 - 2029 National Institutes of Health (R01 AI182355)
Combining B. thuringiensis crystal proteins with small molecule anthelmintics to combat parasitic nematode resistance
PI: Aroian (Univ. of Massachusetts Chan Med School), co-PI: Andersen

PAST

2025 Genetic Resources Core Facility Discovery Initiative Award recipient
Optimization of ultra-long nanopore reads
PI: Andersen

2024 - 2026 National Institutes of Health (R21 AI180805)
Investigating the genetics and genomics of antiparasitic drug responses in a poultry ascarid model
PI: Andersen

2020 - 2026 National Institutes of Health (R01 AI153088) - NIAID
Discovery of novel benzimidazole resistance mechanisms
Lead PI: Andersen, co-PIs: Fraser (UCSF), Gilleard (U Calgary), Kaplan (U Georgia)

2022 - 2025 National Science Foundation, Capacity: Biological Collections (2224885)
Enhancement of the Caenorhabditis Natural Diversity Resource
co-PI: Andersen, co-PI: Tanny

2023 - 2025 US Poultry and Egg Association
Discovery of Novel Anthelmintic Resistance Genes to Reduce Production Loss Caused by Blackhead
PI: Andersen

2022 - 2024 National Institutes of Health (R21 OD030067) - Office of the Director
Genetic and genomic tools for C. briggsae research
Lead PI: Chamberlin (OSU), co-PI: Andersen

2018 - 2024 National Science Foundation CAREER Award (1751035)
Discovery of the molecular mechanisms underlying microevolution of phenotypic plasticity in a developmental trait
PI: Andersen

2018 - 2024 National Institutes of Health (R01 ES029930) - NIEHS
Discovery of conserved molecular mechanisms underlying population-wide variation in toxin responses
Lead PI: Andersen, co-PIs: Baugh (Duke), Rockman (NYU)

2019 - 2024 Human Frontiers Science Program Research Grant (RGP0001/2019)
The repeatability of the genetic mechanisms underlying behavioral evolution
Lead PI: Andersen, co-PIs: Brown (Imperial, MRC), Hodgins (Monash)

2022 - 2025 National Science Foundation, Research Experiences for Undergraduates (2150134)
No funding to REU leadership
REU Site: Quantitative Biology REU (QBREU) at Northwestern University

2018 - 2023 National Science Foundation (1764421) and Simons Foundation (597491)
Research Center for Mathematics of Complex Biological Systems
Understanding organismal growth and development through quantitative approaches

2022 - 2023 Les Turner ALS Center Pilot Grant
Creation of C. elegans ALS models incorporating natural variation

2019 - 2022 National Science Foundation, Collections in Support of Bio. Res. - Living Collections (1930382)
Enhancement of the Caenorhabditis Natural Diversity Resource

2017 - 2021 National Institutes of Health (R01 DK115690) - NIDDK
Large scale nutrigenetics and genomics in a tractable metazoan model

2015 - 2020 American Cancer Society Research Scholar Grant (127313-RSG-15-135-01-DDC)
Elucidating the genetic causes of variation in chemotherapy-based toxicity

2017 - 2020 National Institutes of Heath (R21 AG053638) - NIA
High-throughput multi-modal analysis of natural variation in C. elegans healthspan

2016 - 2019 National Institutes of Heath (R21 AI121836) - NIAID
Discovery and validation of avermectin resistance loci in free-living and parasitic nematodes

2014 - 2019 Pew Charitable Trusts, Scholars Program in the Biomedical Sciences
Elucidating the genetics of anthelmintic resistance in nematode-borne neglected tropical diseases

2015 - 2017 March of Dimes Basil O'Connor Starter Research Grant
Identification of hookworm anthelmintic resistance genes to ameliorate maternal and infant anemia

2014 - 2016 Chicago Biomedical Consortium, Catalyst Grant
Uncovering "missing heritability" in an experimentally tractable model organism

PUBLICATIONS

h-index=49, i10-index=92, link to Google Scholar page (here)
Andersen lab graduate students denoted in *italics* and post-docs denoted in underline.

PREPRINTS:

120. Multiscale learning and topological analysis across complex postures enable robust nematode size quantification in pharmacological assays
bioRxiv, Posted July 7, 2026
DOI: https://doi.org/10.64898/2026.07.12.738008

119. Blevins LJ, Grieve A, Andersen EC, and Kasimatis KR
A male expressed copulatory plug increases female fertility in *Caenorhabditis* nematodes
bioRxiv, Posted Jun. 29, 2026
DOI: https://doi.org/10.64898/2026.06.29.735219

118. Clites BL, Frohock B, Koury EJ, Andersen EC, and Pierce JT (2024)
Natural variation in protein kinase D modifies alcohol sensitivity in *Caenorhabditis elegans*
bioRxiv, Posted Jun. 9, 2024
DOI: https://doi.org/10.1101/2024.06.09.598102

117. Fouad AD, Churgin MA, Hayden J, Xu J, Park JI, Liu A, Teng C, Sun H, Parrado M, Bowlin P, La Torre MD, Crombie TA, Sedore CA, Coleman-Hulbert AL, Johnson E, Phillips P, Andersen EC, and Fang-Yen C (2021)
High-throughput imaging of *Caenorhabditis elegans* aging using collective activity monitoring
bioRxiv, Posted Oct. 19, 2021
DOI: https://doi.org/10.1101/2021.10.18.464905

PEER-REVIEWED:

116. *Moya ND*, Wang B, Tanny RE, Sauria MEG, *O'Connor LM*, Khorshidian A, *McKeown R*, Gosse C, *Dilks CM*, Crombie TA, Zhang G, Rais E, Frezal L, Dan VD, Haryoso E, Devi MP, Gimond C, *Cook DE*, Hsu JC, Shaver AO, *Zdraljevic S*, Richaud A, Wen T, Mehraj A, Sharanya H, Arulprakasam KR, Koury EJ, Roberto NM, *Schaye ES*, Singh V, Tarno H, Ailion M, Paaby AB, Zhao Z, Cutter AD, Wang J, Rockman MV, Felix MA, and Andersen EC
Population structure and punctuated genomic hyper-diversity in *Caenorhabditis briggsae*
Molecular Biology and Evolution. 2026 Aug 1;43(8):msag202.
DOI: https://doi.org/10.1093/molbev/msag202

115. Wang B, *Moya ND*, Tanny RE, Sauria MEG, *O'Connor LM*, Khorshidian A, McKeown R, Stevens L, Buchanan C, Crombie TA, *Dilks CM*, *Evans KS*, *Cook DE*, Zhang G, *Stinson LA*, Roberto NM, Lee D, *Zdraljevic S*, Gosse C, Gimond C, Chen ME, Dang VD, Wang J, Cutter AD, Rockman MV, Felix MA, Braendle C, and Andersen EC
Global genomic diversity of the selfing nematode *Caenorhabditis tropicalis* correlates with geography
Genetics. 2026 Jul 28:iyag196.
DOI: https://doi.org/10.1093/genetics/iyag196

114. Polk EA, Tanny RE, Shaver AO, Andersen EC, Gokcumen O, and Ferkey DM
Natural variation in *C. elegans* 1-octanol olfactory avoidance behavior
G3. 2026 Jul 9:jkag189.
DOI: https://doi.org/10.1093/g3journal/jkag189

113. *O'Connor LM*, *Moya ND*, Jhaveri NS, Tanny RE, Khorshidian A, Lyu H, Chamberlin HM, Baird SE, and Andersen EC
A gap-free, telomere-to-telomere genome assembly for the *Caenorhabditis briggsae* reference strain AF16
G3. 2026 Aug 5;16(8):jkag158.
DOI: https://doi.org/10.1093/g3journal/jkag158

112. *O'Connor LM*, Singh A, Luallen RJ, Zhang G, and Andersen EC
Searching for vertically transmitted endosymbionts in over 4,000 wild strains of three self-fertilizing *Caenorhabditis* species
MicroPublication Biology. 2026 Jun 8: 10.17912/micropub.biology.002151
DOI: https://doi.org/10.17912/micropub.biology.002151

111. Shaver AO, *McKeown R*, Reyes Otero JM, Collins JB, Hogan DW, Fraser JS, Dryer SM, Ragsdale EJ, and Andersen EC (2025)
Evaluating beta-tubulin variants as predictors of benzimidazole resistance across *Caenorhabditis* nematodes
PLoS Pathogens 22(6): e1014306.
DOI: https://doi.org/10.1371/journal.ppat.1014306

110. Saber S, Rifat MMI, Rahimi F, Snyder M, Singh A, Eickwort B, Newhall Y, Rajaei M, Saxena AS, Tanny RE, Katju V, Andersen EC, Zhou J, and Baer CF
Evolution of the rate, spectrum, and fitness effects of mutation under minimal selection in *Caenorhabditis elegans*
Genetics. 2026 Mar 26:iyag085.
DOI: https://doi.org/10.1093/genetics/iyag085

109. Collins JB and Andersen EC
Methods for single-pair *Ascaridia galli* genetic crosses
MicroPublication Biology. 2026 Mar 17: 10.17912/micropub.biology.002043
DOI: https://doi.org/10.17912/micropub.biology.002043

108. Crombie TA, *McKeown R*, Widmayer SJ, Shaver AO, *Moya ND*, Collins JB, Wit J, Tanny RE, Braendle C, Stevens L, van Sluijs L, Rockman MV, Sterken MG, Felix MA, and Andersen EC
Natural variation suggests candidate genes underlying *Caenorhabditis elegans* susceptibility to diverse toxicants
Toxicological Sciences. 2026 Mar. 2;209(3):kfag019.
DOI: https://doi.org/10.1093/toxsci/kfag019

107. Wei X, Richaud A, Tanny RE, Andersen EC, and Felix MA
Mutational divergence over years in local populations of the selfing nematode *Caenorhabditis elegans*
Genetics. 2026 Feb. 17: iyag043.
DOI: https://doi.org/10.1093/genetics/iyag043

106. Crombie TA, Pamminger T, Andersen EC, and Glaberman S (2026)
High-Throughput Toxicity Screening with *C. elegans*: Current Platforms, Key Advantages, and Future Directions
Environmental Science and Technology. 2026 Jan 20;60(2):1560-1576.
DOI: https://doi.org/10.1021/acs.est.5c12562

105. Peake AL, Jhaveri NS, Andersen EC, and Stinchcombe JR (2025)
High-throughput developmental assay of cold tolerance in *Caenorhabditis elegans*.
MicroPublication Biology. 2025 Nov 5: 10.17912/micropub.biology.001850.
DOI: https://doi.org/10.17912/micropub.biology.001356

104. Polk EA, Aqil A, Collins JB, Tanny RE, Andersen EC, Gokcumen O, and Ferkey DM (2025)
Wild Strains Reveal Natural Variation in *C. elegans* Avoidance Behaviors.
G3 (Bethesda). 2025 Oct 10:jkaf243.
DOI: https://doi.org/10.1093/g3journal/jkaf243

103. Collins JB, Choo R, Shaver AO, *Schaye ES*, Volpe T, Nunn L, Lighty ME, Niel KR, Frye EM, Zamanian M, and Andersen EC (2025)
A small-scale survey of fenbendazole resistance in *Ascaridia galli* and *Heterakis gallinarum*, two common ascarid parasites of poultry.
Poultry Science 2025 Sep 10;104(11):105808
DOI: https://doi.org/10.1016/j.psj.2025.105808

102. Jhaveri NS, *Mastronardo MK*, Collins JB, and Andersen EC (2025)
Development of a size-separation technique to isolate *Caenorhabditis elegans* embryos using mesh filters
PLoS ONE, 2025 Apr 24;20(4):e0318143
DOI: https://doi.org/10.1371/journal.pone.0318143

101. Jhaveri NS and Andersen EC (2025)
Fecundities of Hawaiian *Caenorhabditis briggsae* wild strains are not correlated with natural niche temperatures.
MicroPublication Biology, 2025 Jan 15; 2025:10.17912/micropub.biology.001356
DOI: https://doi.org/10.17912/micropub.biology.001356

100. Collins JB, *Dilks CM*, Hahnel SR, Rodriguez B, Fox BW, Redman E, Yu J, Cooke B, Sihuta K, Zamanian M, Roy PJ, Schroeder FC, Gillard JS, and Andersen EC (2024)
Naturally occurring variation in a cytochrome P450 modifies thiabendazole responses independent of beta-tubulin
PLoS Pathogens, 2025 Jan 14;21(1):e1012602
DOI: https://doi.org/10.1371/journal.ppat.1012602

99. *Moya ND*, Yan SM, McCoy RC, and Andersen EC (2024)
The long and short of hyper-divergent regions
Trends in Genetics, 2024 Nov 15
DOI: https://doi.org/10.1016/j.tig.2024.11.005

98. Shaver AO and Andersen EC (2024)
Integrating metabolomics into the diagnosis and investigation of anthelmintic resistance
Trends in Parasitology, 2024 Dec;40(12):1097-1106.
DOI: https://doi.org/10.1016/j.pt.2024.10.020

97. Zhang G, Felix MA, and Andersen EC (2024)
Transposon-mediated genic rearrangements underlie variation in small RNA pathways
Science Advances, 2024 Sep 20;10(38):eado9461.
DOI: https://doi.org/10.1126/sciadv.ado9461

96. Collins JB, Shaver AO, Schaye ES, Volpe T, Nunn LR, Zamanian M, and Andersen EC (2024)
Lack of correlation between parasite burden and key weight metrics in poultry infected with intestinal ascarids
MicroPublication Biology, 2024 Aug 7;2024:10.17912/micropub.biology.001197.
DOI: https://doi.org/10.17912/micropub.biology.001197

95. Crombie TA, Raja M, Saxena AS, Johnson LM, Saber S, Tanny RE, Ponciano JM, Andersen EC, Zhou J, and Baer CF (2024)
Direct inference of the distribution of fitness effects of spontaneous mutations from recombinant inbred *C. elegans* mutation accumulation lines
Genetics, 2024 Aug 14:iyae136.
DOI: https://doi.org/10.1093/genetics/iyae136

94. Collins JB, Stone S, Koury E, Paredes A, Shao F, Lovato C, Chen M, Shi R, Li A, Candal I, Al Moutaa K, *Moya ND*, and Andersen EC (2024)
Quantitative tests of albendazole resistance in beta-tubulin mutants
International Journal for Parasitology: Drugs and Drug Resistance, 2024 Jul 7;25(100556),
DOI: https://doi.org/10.1016/j.ijpddr.2024.100556

93. Yang H, Lee D, Kim H, *Cook DE*, Paik YK, Andersen EC, and Lee J (2024)
Glial expression of a steroidogenic enzyme underlies natural variation in hitchhiking behavior
PNAS 2024 Jul 3;121(28)e2320796121, DOI: https://www.pnas.org/doi/10.1073/pnas.2320796121

92. Knox J, Burns AR, Cooke B, Cammalleri SR, Kitner M, Ching J, Castelli JMP, Puumala E, Snider J, Koury E, Collins JB, Andersen EC, Stagljar I, Cowen LE, Laurens M, Zasada I, and Roy PJ (2024)
Cyprocide selectively kills nematodes via cytochrome P450 bioactivation
Nature Communications 2024 July 2;15(5529), DOI: https://doi.org/10.1038/s41467-024-49738-4

91. Shaver AO, Miller IR, Schaye ES, *Moya ND*, Collins JB, Wit J, Blanco AH, Shao FM, Andersen EJ, Khan SA, Paredes G, and Andersen EC (2024)
Quantifying the fitness effects of resistance alleles with and without anthelmintic selection pressure using *C. elegans*
PLoS Pathogens 2024 May 20;20(5):e1012245, DOI: https://doi.org/10.1371/journal.ppat.1012245

90. Lee H, Boor SA, Hilbert ZA, Meisel JD, Park J, *Wang Y*, *McKeown R*, Fischer SEJ, Andersen EC, and Kim DH (2023)
Genetic variants that modify the neuroendocrine regulation of foraging behavior in *C. elegans*
Science Advances 2024 Jun 14;10(20):eadk9481, DOI: https://doi.org/10.1126/sciadv.adk9481

89. Iyu H, *Moya ND*, Andersen EC, and Chamberlin HM (2024)
Gene duplication and evolutionary plasticity of *lin-12*/Notch gene function in *Caenorhabditis*
Genetics 2024 May 29:iyae064
DOI: https://doi.org/10.1093/genetics/iyae064

88. Wolstenholme AJ, Andersen EC, Choudhary S, Ebner F, Hartmann S, Holden-Dye L, Kashyap SS, Krücken J, Martin RJ, Midha A, Neysum P, Neveu C, Robertson AP, von Samson-Himmelstjerna G, Walker R, Wang J, Whitehead BJ, and Williams PDE
Getting around the roundworms: Identifying knowledge gaps and research priorities for the ascarids
Advances in Parasitology 2024 Feb 20
DOI: https://doi.org/10.1016/bs.apar.2023.12.002

87. Crombie TA, *McKeown R*, *Moya ND*, *Evans KS*, Widmayer S, LaGrassa V, Roman N, Tursunova O, Zhang G, Gibson S, Buchanan C, Roberto N, Vieira R, Tanny RE, and Andersen EC
CaeNDR, the *Caenorhabditis* Natural Diversity Resource
Nucleic Acids Research 2023 Oct 19; gkad887
DOI: https://doi.org/10.1093/nar/gkad887

86. *Moya ND*, Stevens L, Miller IR, Galindo JL, Bardas AD, Yeo C, Rozenich AJ, Xu M, Koh ESH, and Andersen EC (2023)
Novel and improved *Caenorhabditis briggsae* gene models generated by community curation
BMC Genomics 2023 Aug 25;24(1):486
DOI: https://doi.org/10.1186/s12864-023-09582-0

85. Wit J, *Dilks CM*, Zhang G, Kim Guisbert KS, *Zdraljevic S*, Guisbert E, and Andersen EC (2023)
Praziquantel inhibits *Caenorhabditis elegans* development and species-wide differences might be *cct-8*-dependent
PLoS ONE 2023 Aug 10;18(8):e0286473
DOI: http://doi.org/10.1371/journal.pone.0286473

84. Lee D, Fox BW, Palomino DCF, Panda O, Tenjo FJ, *Evans KS*, Stevens L, Koury E, Rodrigues PR, Kolodziej AR, Schroeder KF, and Andersen EC (2023)
Natural genetic variation in the pheromone production of *C. elegans*
The Proceedings of the National Academy of Sciences 2023 May 10;120(26)e2221150120
DOI: https://doi.org/10.1073/pnas.2221150120

83. Zhang G and Andersen EC (2023)
Interplay between polymorphic short tandem repeats and gene expression variation in *Caenorhabditis elegans*
Molecular Biology and Evolution 2023 Apr 4;40(4):msad067
DOI: https://doi.org/10.1093/molbev/msad067

82. Shaver AO, Wit J, *Dilks CM*, Crombie TA, Li H, Aroian RV, and Andersen EC (2023)
Variation in anthelmintic responses are driven by genetic differences among diverse *C. elegans* wild strains
PLoS Pathogens 2023 Apr 3;19(4):e1011285
DOI: https://doi.org/10.1371/journal.ppat.1011285

81. Venkatesan A, Jimenez Castro PD, Morosetti A, Horvath H, Chen R, Redman E, Dunn K, Collins JB, Fraser JS, Andersen EC, Kaplan RM, and Gilleard JS (2023)
Molecular evidence of widespread benzimidazole drug resistance in *Ancylostoma caninum* from domestic dogs throughout the USA and discovery of a novel β-tubulin benzimidazole resistance mutation
PLoS Pathogens 2023 Mar 2;19(3):e1011146
DOI: https://doi.org/10.1371/journal.ppat.1011146

80. Collins JB and Andersen EC (2022)
The turkey ascarid, *Ascaridia dissimilis*, as a model genetic system
International Journal for Parasitology 2022 Dec 19;S0020-7519(22)00177-1
DOI: https://doi.org/10.1016/j.ijpara.2022.10.005

79. Lesack K, Mariene GM, Andersen EC, and Wasmuth JD (2022)
Different structural variant prediction tools yield considerably different results in *Caenorhabditis elegans*
PLoS ONE 2022 Dec 30;17(12):e0278424
DOI: https://doi.org/10.1371/journal.pone.0278424

78. *Nyaanga J*, Shirman S, Mangan NM, and Andersen EC (2022)
Characterization of larval growth in *C. elegans* cuticle mutants
microPublication Biology 2022 Nov. 4, 2022:10.17912/micropub.biology.000662
DOI: https://doi.org/10.17912/micropub.biology.000662

77. Gibson S, Ness-Cohn E, and Andersen EC (2022)
Benzimidazoles cause lethality by inhibiting the function of *Caenorhabditis elegans* neuronal beta-tubulin
International Journal for Parasitology: Drugs and Drug Resistance 2022 Dec;20:89-96
DOI: https://doi.org/10.1016/j.ijpddr.2022.10.004

76. Shaver AO, Garcia BM, Gouveia GJ, Morse AM, Liu Z, Asef CK, Borges RM, Leach FE, Andersen EC, Amster IJ, Fernadez FM, Edison AS, and McIntyre LM (2022)
An anchored experimental design and meta-analysis approach to address batch effects in large-scale metabolomics
Frontiers Molecular Biosciences 2022 Nov 9;9:930204
DOI: https://doi.org/10.3389/fmolb.2022.930204

75. Zhang G, *Wang Y*, and Andersen EC (2022)
Natural variation in *C. elegans* short tandem repeats
Genome Research 2022 Oct;32(10):1852-1861
DOI: https://doi.org/10.1101/gr.277067.122

74. Widmayer SJ, Crombie TA, *Nyaanga JN*, *Evans KS*, and Andersen EC (2022)
C. elegans toxicant responses vary among genetically diverse individuals
Toxicology 2022 Aug 20; 479:153292
DOI: https://doi.org/10.1016/j.tox.2022.153292

73. *Nyaanga J* and Andersen EC (2022)
Linkage mapping reveals loci that underlie differences in *C. elegans* growth
G3 2022 Sep 30;12(10):jkac207
DOI: https://doi.org/10.1093/g3journal/jkac207

72. Wit J, Workentine ML, Redman E, Laing R, Stevens L, Cotton JA, Chaudry U, Ali Q, Andersen EC, Yeaman S, Wasmuth JD, and Gilleard JS (2022)
Genomic signatures of selection associated with benzimidazole drug treatments in *Haemonchus contortus* field populations
International Journal for Parasitology: Drugs and Drug Resistance 2022 Sep;52(10):677-689
DOI: https://doi.org/10.1016/j.ijpara.2022.07.004

71. Webster AK, Chitrakar R, Powell M, Chen J, Fisher K, Tanny RE, Stevens L, *Evans KS*, Antoshechkin I, Andersen EC, and Baugh LR
Using population selection and sequencing to characterize natural variation of starvation resistance in *Caenorhabditis elegans*
eLife 2022 Jun 21; 11:e80204
DOI: https://doi.org/10.7554/eLife.80204

70. Zhang G, Roberto NM, Lee D, Hahnel SR, and Andersen EC (2022)
The impact of species-wide gene expression variation on *Caenorhabditis elegans* complex traits
Nature Communications 2022 Jun 16; 13(1):3462
DOI: https://doi.org/10.1038/s41467-022-31208-4

69. Pallotto LM, *Dilks CM*, Park YJ, Smit RB, Lu B, Gopalakrishnan C, Gilleard JS, Andersen EC, and Mains PE (2022)
Interactions of *Caenorhabditis elegans* β-tubulins with the microtubule inhibitor and anthelmintic drug albendazole
Genetics 2022 Jul 30; 221(4):iyac093
DOI: https://doi.org/10.1093/genetics/iyac093

68. Fox BW, Ponomarova O, Lee YU, Zhang G, Giese GE, Walker M, Roberto NM, Na H, Reis-Rodriguez P, Curtis BJ, Kolodziej AR, Crombie TA, *Zdraljevic S*, Yilmaz LS, Andersen EC, Schroeder FC, and Walhout AJM (2022)
C. elegans as a model for inter-individual variation in metabolism
Nature 2022 Jul; 607(7919):571-577
DOI: https://doi.org/10.1038/s41586-022-04951-3

67. Crombie TA, Chikuturudzi C, *Cook DE*, and Andersen EC (2022)
An automated approach to quantify chemotaxis index in *C. elegans*
microPublication Biology 2022 May 26; 2022:10.17912/micropub.biology.000567
DOI: https://doi.org/10.17912/micropub.biology.000567

66. Widmayer SJ, *Evans KS*, *Zdraljevic S*, and Andersen EC (2021)
Evaluating the power and limitations of genome-wide association mapping in *C. elegans*
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Publications from research prior to my first tenure-track position:

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PROFESSIONAL SEMINARS

Departmental seminars and invited conference presentations (not including trainees):

2027 Dept. of Infectious Disease, Univ. of Georgia, Athens, GA
Dept. of Biology, Georgia Tech University, Atlanta, GA
Dept. of Molecular and Cell Biology, Univ. of California - Berkeley, Berkeley, CA
Parasitic Helminths: New Perspectives in Biology and Infection, Hydra, Greece (keynote, scheduled)

2026 American Society of Tropical Medicine and Hygiene, Washington, DC (workshop keynote)
Institute for Parasitology, McGill University, Montreal, Canada
Dept. of Microbiology, Immunology, and Tropical Medicine, George Washington Univ., Washington, DC
Bronx High School of Science, New York NY (outreach lecture)
16th International Conference of Parasitology, Montreal, Canada (two invited talks)
Society of Nematologist, Baltimore, MD (keynote)
AAVP, Pre-meeting: Optimizing Parasiticides in the Face of Drug resistance (organizer, speaker)
European Society of Nematologists, Egmond can Zee, Netherlands (workshop keynote)
Dept. of Biology, Univ. of Maryland, College Park, MD

2025 Dept. of Biology, Univ. of Buffalo, Buffalo, NY
EMBL Workshop on Wild Frontiers of Model Organisms, Heidelberg, Germany
Les Treilles Foundation meeting on nematode evolution, Nice, France

2024 Molecular genetics, genomics, microbiology, University of Strasbourg, Strasbourg, France
Dept. of Biology, Lehigh University, Lehigh, PA
Dept. of Biochemistry, Cellular, and Molecular Biology, Univ. of Tennessee, Knoxville, TN
Dept. of Human Genetics, University of California - Los Angeles
Evolution of *Caenorhabditis* and other nematodes, Vienna, Austria
Poultry Science Dept., Univ. of Veterinary Medicine - Vienna, Vienna, Austria
NIH, Laboratory of Parasitic Diseases, Bethesda, MD
Carnegie Institute of Embryology, Baltimore, MD
Baltimore Worm Club

2023 Chicago Area Worm Meeting (keynote)
Zoom about Parasites (ZaP worms), virtual seminar
Latin American Worm Meeting, Valparaiso, Chile
2022 Institute for Integrative Genome Biology, Univ. of California - Riverside, Riverside, CA
International Congress of Toxicology, Maastricht, Netherlands
Human Frontiers Research Symposium, Paris, France
The *C. elegans* Metabolism, Aging, Pathogenesis, Stress and Small RNAs
Genetics and Molecular Biology seminar series, University of North Carolina - Chapel Hill
Dept. of Biology, Oklahoma University, Norman, OK
Wellcome Evolutionary Systems Biology
2021 The Ascarid Research and Training Initiative, Lyon, France (Keynote)
Bridging the Divide: International Worm Meeting (Keynote)
66th Annual Meeting of the American Association of Veterinary Parasitologists (Keynote)
Max Planck Institute for Evolutionary Biology, Plön, Germay
Institut de Biologie Valrose, Nice, France
2020 American Society of Tropical Medicine and Hygiene, symposium on genetic crosses
Dept. of Molecular Medicine, Univ. of Massachusetts Medical School, Worcester, MA
Oakton Community College, Des Plaines, IL
Chengdu Research Base of Giant Panda Breeding, Chengdu, People's Republic of China
2019 Netherlands Institute for Ecology (NIOO-KNAW), Wageningen, Netherlands
Dept. of Genetics, Wageningen Univ. and Research, Wageningen, Netherlands
Dept. of Nematology, Wageningen Univ. and Research, Wageningen, Netherlands
Dept. of Genome Sciences, Univ. of Washington, Seattle, WA
Institute of Molecular Biology, Academia Sinica, Taiwan
Dept. of Genetics, Univ. of Georgia, Athens, GA
Dept. of Infectious Diseases, Univ. of Georgia, Athens, GA
Dept. of Biology, Georgia Institute of Technology, Atlanta, GA
Ingram Cancer Center, Vanderbilt University, Nashville, TN
2018 Max Planck Institute for Developmental Biology, Tübingen, Germany
Dept. of Evolutionary Ecology and Genetics, Christian-Albrechts-Universitat, Kiel, Germany
Berlin Seminar for Resistance Research, Freie Universitat Berlin, Berlin, Germany
Robert H Lurie Comprehensive Cancer Center, Northwestern University, Chicago, IL
Dept. of Genetics, Washington University School of Medicine, St. Louis, MO
Dept. of Genetics, University of Pennsylvania, Philadelphia, PA
Dept. of Pathobiology, University of Pennsylvania, Philadelphia, PA
Dept. of Microbiology, Seoul National University, Seoul, Korea
Dept. of Biology, University of Oregon, Eugene, OR
Dept. of Biological Sciences, University of Southern California, Los Angeles, CA
Dept. of Cell Biology and Anatomy, Rosalind Franklin University, Chicago, IL
New York Univ. Center for Genomics and Sys. Bio. Parasite Workshop, Abu Dhabi, UAE
Department of Biology, Univ. of California - San Diego, San Diego, CA
Department of Biology, Duke University, Durham, NC
Donnelly Centre for Cellular And Biomolecular Research, University of Toronto, Toronto, ON
Department of Molecular Biology and Genetics, Cornell University, Ithaca, NY
Medical Research Council, London Institute of Medical Sciences, London, UK
Department of Biology, Carnegie Mellon University, Pittsburgh, PA
Host-Parasite Interactions, University of Calgary, Banff, Canada
Florida Area Worm Meeting (Keynote), Florida Institute of Technology, Melbourne, FL
Department of Biology, Skirball Institute, New York University Medical School, New York, NY
Department of Biology, University of Minnesota, Minneapolis, MN
2017 Department of Biology, Indiana University, Bloomington, IN
New York University Center for Genomics and Systems Biology Symposium in Abu Dhabi, UAE

2016 Midwest Quantitative Biology at Purdue University, West Lafayette, IN
Molecular and Cellular Biology of Helminth Parasites X, Hydra, Greece
Computational Research Day, Northwestern University, Evanston, IL
Evolutionary Biology of *Caenorhabditis* and other nematodes (Keynote), CSHL, Cold Spring Harbor, NY
Department of Genetics, University of Utah, Salt Lake City, UT
Department of Biology, University of Iowa, Iowa City, IA
Department of Biomedical Sciences, Iowa State University, Ames, IA
Anthelmintics: Discovery to Resistance II, San Diego, CA
Program in Systems Biology, University of Massachusetts Medical School, Worcester, MA
2015 Evolution seminar series, University of Wisconsin, Madison, WI
Biotechnology Training Program, Northwestern University, Evanston, IL
Department of Biology, Johns Hopkins University, Baltimore, MD
Department of Biology, University of Maryland, College Park, MD
Department of Pharmacology, Feinberg School of Medicine, Northwestern University, Chicago, IL
Midwest Neglected Infectious Disease Meeting, Notre Dame University, South Bend, IN
Quantitative genetics workshop, 20th International *C. elegans* meeting, UCLA, Los Angeles, CA
Michigan Area Worm Meeting, van Andel Institute, Grand Rapids, MI
2014 Northwestern Institute on Complex systems, Northwestern University, Evanston, IL
Fondation de Treilles: Revisiting the roles of phenotypic plasticity in evolution, Provence, France
Biology Department, Marquette University, Milwaukee, WI
Pharmacogenomics group, University of Chicago, Chicago, IL

Seminars before starting at first faculty position:
2013 Quantitative genetics workshop, 19th International *C. elegans* meeting, UCLA, Los Angeles, CA
Molecular Bioscience Department, Northwestern University, Evanston, IL
Program in Systems Biology, University of Massachusetts Medical School, Worcester, MA
2012 Biology Department, Dartmouth University, Hanover, NH
Human Genetics Department and Life Sciences Institute, University of Michigan, Ann Arbor, MI
Genetics Department, University of Georgia, Athens, GA
Biology Department, Case Western Reserve University, Cleveland, OH
Biology Department and BioDesign Institute, Arizona State University, Phoenix, AZ
Center for Computational and Integrated Biology, Rutgers University, Camden, NJ
Biology Department, University of Florida, Gainesville, FL
2011 Evolution workshop, 18th International *C. elegans* meeting, UCLA, Los Angeles, CA
Laboratory of Toxicology, NIEHS, Research Triangle Park, NC
2010 Institute for Evolutionary Biology Department, University of Edinburgh, Edinburgh, UK
2008 Featured talk at *C. elegans* Aging, Stress, and Pathogenesis meeting, Madison, WI
2000 Undergraduate research symposium, Stanford University, Stanford, CA

PEER REVIEW AND RELATED ACTIVITIES

Editorial Board:
2022 - 2024 *Journal of Nematology*
2020 - 2023 *Genes*
2015 - 2026 *Trends in Genetics*

Associate Editor:
2017 - 2019 *BMC Genomics (Multicellular invertebrate genomics)*

Guest Associate Editor:
PLoS Genetics, *eLife*, *PLoS Pathogens*, *PLoS Neglected Tropical Disease*, *Genetics*, *G3*

Reviewing activity: Grants and fellowships

2026 *Ad hoc* reviewer ZRG1 BCTA-B(55)
Ad hoc reviewer ZRG1 BCTA-B(57)
University of California's Early Career Award Reviewer
NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
NSF grant reviewer
Ad hoc reviewer NIH RM1 review panel (February)
Ad hoc reviewer Israeli Science Foundation
Ad hoc reviewer NIH RM1 review panel (January)

2025 NSF grant reviewer
Ad hoc reviewer BBSRC UK
Ad hoc reviewer European Science Foundation
Ad hoc reviewer Swiss National Science Foundation
Ad hoc reviewer NIH Neurotoxicology and Alcohol (NAL) Study Section
Ad hoc reviewer NIH Pathogenic Eukaryotes (PTHE) Study Section

2024 NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Ad hoc reviewer Fulbright Fellowship
International Research Olympiad Advisory Board
Ad hoc reviewer Japan Society for the Promotion of Science
Ad hoc reviewer BBSRC UK
Ad hoc reviewer Swiss National Science Foundation

2023 *Ad hoc* reviewer Dept. of Defense - Neurotoxicity
NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Ad hoc reviewer NIH ViCTER Award
Panel reviewer NSF EDGE
Ad hoc reviewer NIH NIDA Avenir Award
Ad hoc reviewer Dept. of Defense - Neurotoxicity

2022 NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Ad hoc BBSRC grant reviewer
Katholieke Universiteit Leuven grant reviewer
Ad hoc reviewer National Science Foundation (IOS)
Ad hoc reviewer NIH NIDA Avenir Award
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)

2021 ERC Consolidator Grant reviewer (COI, declined)
NIH F30/F31/F32 fellowship Cell, Development, and Systems Biology review panel ZRG F05-Q
Canada Foundation for Innovation reviewer
NSF GRFP reviewer
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)

2020 NIH Special Emphasis panel ZRG IFCN-C (02)
NIH Special Emphasis panel ZRG ETTN-N (02)
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)

2019 *Ad hoc* reviewer for Wellcome Trust Early Career Grant
Panel reviewer on NASA, Flight and Ground Space Biology
Ad hoc reviewer for Agence Nationale de la Recherche
Ad hoc reviewer for Univ. of Wisconsin - Milwaukee Catalyst grant
Panel reviewer on NIH NIAID R13
Ad hoc reviewer National Science Foundation (CAREER)
Ad hoc reviewer for Swiss 3R Competence Centre
Ad hoc reviewer for Austrian Science Foundation
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)

2018 *Ad hoc* reviewer for Alzheimer's Society
Ad hoc reviewer National Science Foundation (IOS)

2018 *Ad hoc* reviewer National Science Foundation (CAREER)
Scientific Advisory Board for P01AI127338 (PI Michael Ferdig, Notre Dame Univ.)
Ad hoc reviewer for Bill and Melinda Gates Foundation
2016 ERC COST grant reviewer
2015 *Ad hoc* reviewer for National Toxicity Program, project assessment
2014 *Ad hoc* reviewer for Human Frontiers Science Program
Ad hoc reviewer for National Science Foundation (IOS)

PROFESSIONAL AFFILIATIONS AND SERVICE

Membership in Professional Societies:
Genetics Society of America
Society of Molecular Biology and Evolution (lifetime)
Society for Evolution (lifetime)
Society for Integrative and Comparative Biology
World Association for the Advancement of Veterinary Parasitology
American Association of Veterinary Parasitologists
Society of Nematologists

Mentorship and Diversity training:
2024 Johns Hopkins University, Center for the Improvement of Mentored Experiences in Research (CIMER), virtual
- Building Trust Through Effective Communication
- Creating a Culture of Inclusion in your Lab
- Aligning Expectations to Promote Independence

2022 Univ. of Wisconsin - Madison, Center for the Improvement of Mentored Experiences in Research (CIMER), in-person
Arizona State Univ., Culturally Aware Mentoring, virtual

2021 Univ. of Southern California, Strategies for Equity-based Holistic Review in Graduate Admissions
Northwestern Univ., Diversity, equity, and inclusion in hiring decisions
2020 NIH OITE training on Health and Wellness of Trainees

Professional service:
2027 International Worm Meeting, elected organizer, Kobe, Japan
Anthelmintics VII Meeting, organizing committee, Queens University, Belfast
AAVP, Parasite Resource Naming Committee Chair
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
2026 AAVP, Pre-meeting workshop co-organizer
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
2025 International Worm Meeting, organizing committee/abstract reviewer
2024 External Departmental Review Committee, Department of Biology, University of Pennsylvania
C. elegans and other nematodes meeting organizer, IMBA, Vienna, Austria
External thesis examiner member for Victor Loegler (Schacherer lab, Univ. of Strasbourg
C. elegans Community Faculty Mentor, Genetics Society of America
External thesis examiner member for Maria Mercado (Fraser lab, Univ. of Toronto)
External thesis committee member for Youn Jae Kang (Ding lab, Max Planck Institute of Animal Behavior)

2023 External thesis examiner for Nikita Jhaveri (Gupta lab, McMaster Univ.)
Worm Board *ex officio* member Nematode Genomes
C. elegans Community Faculty Mentor, Genetics Society of America
External thesis committee member for Youn Jae Kang (Ding lab, Max Planck Institute of Animal Behavior)

2022 *C. elegans* Community Faculty Mentor, Genetics Society of America
External thesis committee member for Leonor Gianechini (Moorhead lab, Univ. of Georgia)
External thesis committee member for Youn Jae Kang (Ding lab, Max Planck Institute of Animal Behavior)
External thesis committee member for Amanda Shaver (Edison lab, Univ. of Georgia)
Worm Board *ex officio* member Nematode Genomes
Organizing committee Anthelmintics V: Resistance and vaccines
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)

2021 Worm Board *ex officio* member Nematode Genomes
Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)

2020 Worm Board *ex officio* member Nematode Genomes
Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)

2019 External thesis committee examiner for Yiru Wang (Kammenga lab, Wageningen Univ. and Research)
External thesis committee member for Amanda Shaver (Edison lab, Univ. of Georgia)
External thesis committee examiner for Aurian Garcia-Gonzalez (Walhout lab, UMMS)
Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)

2018 Organizing committee for *C. elegans* dev., cell bio., and gene exp. meeting (Barcelona, Spain)
Chair of the Natural Variation session, *C. elegans* dev., cell bio., and gene exp. meeting
Co-organizer of the Chicago Area Worm Meeting (ChAWM, www.chawm.org)
Scientific Advisory Board Member for P01AI127338 (PI Dr. Michael Ferdig, Notre Dame Univ.)
Organizing committee for Parasitic Nematodes meeting, NYU Abu Dhabi
External thesis committee member for Victoria Vu (Fraser lab, University of Toronto)
Poster judge, Northwestern Undergraduate Research Symposium

2017 Organizing committee for the 21st International *C. elegans* meeting
Chair of the Evolution and Ecology parallel session, 21st International *C. elegans* meeting

2015 Organizing committee for the 20th International *C. elegans* meeting
Poster judge, 20th International *C. elegans* meeting - Evolution and Genomics section
Genetics Soc. of America Mentor Lunch, *Postdoc search*, 20th International *C. elegans* meeting

2014 Panelist, NUIN Post-doc Association, *Interviews and Start-up packages*
Poster judge, Northwestern Undergraduate Research Symposium
Panelist, Pathways to the Professoriate, *How to prepare for a job interview?*

2013 Poster judge, Northwestern Undergraduate Research Symposium
Panelist, Bioscientist Freshman seminar; *How to find a research lab?*
Poster judge, 19th International *C. elegans* meeting - Evolution and Genomics section

***C. elegans* community service and open-data:**

2016 - 2021 Creator and director of the *C. elegans* Natural Diversity Resource (CeNDR)
This resource organizes and disseminates wild *C. elegans* strains, whole-genome sequence data, and enables genome-wide association mappings through a cloud-based service.

2021 - Creator and director of the *Caenorhabditis* Natural Diversity Resource (CaeNDR, *link*)
This resource expands on the on CeNDR after incorporating wild strains from *C. briggsae* and *C. tropicalis.* We also added new web tools and analysis frameworks to enable comparative and molecular quantitative genetics. All data are freely available to all.

TEACHING AND ADVISING

Undergraduate teaching:

2026 AS.020.383: *Advanced Genetics* (fall semester, 30 students)
AS.020.303: *Genetics* (summer semester, 11 students)
AS.020.503: *Independent study in Biology* (fall and spring semesters, Rohan Shrishrimal, Kate Stone, Leah Froemming)

2025 **New course: AS.020.383: *Advanced Genetics* (fall semester, 23 students)**
AS.020.314: *Biology of Disease* (fall semester, 180 students)
AS.020.503: *Independent study in Biology* (fall and spring semesters, Rohan Shrishrimal, Kate Stone)

2024 AS.020.303: *Genetics* (spring semester, 465 students)
AS.020.314: *Biology of Disease* (fall semester, 180 students)
AS.020.503: *Independent study in Biology* (fall and spring semesters, Rohan Shrishrimal, Kate Stone)

2023 Biological Sciences 203: *Genetics and Evolution* (winter, 285 students)
Biological Sciences 398: *Independent Research* (Fiona Shao, Alyssa Blanco)
Biological Sciences 399: *Independent Research* (Andrea Phung, Cassia Yeo, Sharik Khan)

2022 **New course: Biological Sciences 203: *Genetics and Evolution* (winter, 379 students)**
Biological Sciences 398: *Independent Research* (Andrea Phung)

2021 **Biological Sciences 393: *Genetic Analysis* (spring, 31 students)**
Biological Sciences 399: *Independent Research* (Karan Gowda, Kailyn Parham, Katie Introcaso)

2020 **Biological Sciences 215: *Genetics and Molecular Biology* (spring, 492 students)**
Biological Sciences 399: *Independent Research* (Anna Derrick, Emily Jahn, Jake Mostad, Kailyn Parham)

2019 **Biological Sciences 393: *Biomedical Genetics* (spring, 19 students)**
Biological Sciences 398: *Tutorial in Biology* (Anna Derrick, Emily Jahn, Kailyn Parham)
Biological Sciences 399: *Independent Research* (Karol Bisaga, Grace Park, Jake Mostad)

2018 **Biological Sciences 393: *Genetic Analysis* (winter, 28 students)**
Biological Sciences 398: *Tutorial in Biology* (Karol Bisaga)
Biological Sciences 399: *Independent Research* (Karol Bisaga, Kimberly Collins, Selina Deiparine, Grace Park)

2017 **Biological Sciences 393: *Genetic Analysis* (spring, 22 students)**
Biological Sciences 398: *Tutorial in Biology* (Kimberly Collins)
Biological Sciences 399: *Independent Research* (Selina Deiparine, Samuel Hamilton, Grace Park)

2016 **Biological Sciences 393: *Genetic Analysis* (spring, 17 students)**
Biological Sciences 398: *Tutorial in Biology* (Sarah Bier, Mattlyn Cordova, Selina Deiparine, Samuel Hamilton, Grace Park)

2015 *Guest Lecture:* University of Wisconsin-Madison Biology 675 - *Evolution seminar* (fall, 8 students)
New course: Biological Sciences 393: *Genetic Analysis* (spring, 10 students)
Biological Sciences 398: *Tutorial in Biology* (Lautaro Cilenti)
Biological Sciences 399: *Independent Research* (Kreena Patel, Hillary Tsang)

2014 Biological Sciences 398: *Tutorial in Biology* (Mazeed Aro-Lambo, Kreena Patel, Hillary Tsang)

Graduate teaching:

2026 AS.020.683: *Advanced Genetics* (fall semester, 5 students)

2025 **New course: AS.020.683: *Advanced Genetics* (fall semester, 22 students)**
CMDB RCR course - *Collaborative Science* (one lecture, 26 students)

2024 CMDB RCR course - *Collaborative Science* (one lecture, 22 students)

2022 Driskill Graduate Program 430: Genetic Analysis - Leveraging Big Data and Model Organisms (spring, one guest lecture, 12 students)

2021 **New course:** Interdisciplinary Biological Sciences: *Introduction to R for biologists* (fall, Five students)
Driskill Graduate Program 430: Genetic Analysis - Leveraging Big Data and Model Organisms (spring, one guest lecture, 16 students)

2020 Interdisciplinary Biological Sciences 423: *Ethics of collaboration* (fall, one guest lecture, 45 students)

2019 Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility* (summer, one guest lecture, 17 students)

2018 Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility* (summer, one guest lecture, 25 students)
Interdisciplinary Biological Sciences/Chemistry 416: *Practical Training in Chemical Biology Methods and Experimental Design* (spring, five lectures, 10 students)

2017 Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility* (summer, one guest lecture, 15 students)
Interdisciplinary Biological Sciences/Chemistry 416: *Practical Training in Chemical Biology Methods and Experimental Design* (spring, five lectures, 8 students)
Interdisciplinary Biological Sciences 421: *Rigor and Reproducibility* (spring, one guest lecture, 5 students)

2016 Interdisciplinary Biological Sciences: *Graduate Computational Biology Bootcamp* (fall, 3 days, eight hours per day, 19 students) - www.GitHub.com/AndersenLab/IBiS-Bootcamp

2015 Interdisciplinary Biological Sciences 402: *Eukaryotic Molecular Biology* (fall, one guest lecture, 22 students)
Interdisciplinary Biological Sciences: *Graduate Computational Biology Bootcamp* (fall, 3 days, eight hours per day, 22 students) - www.GitHub.com/AndersenLab/IBiS-Bootcamp
Interdisciplinary Biological Sciences 423: *Ethics of peer review* (spring, one guest lecture, 41 students)

2014 Interdisciplinary Biological Sciences 402: *Eukaryotic Molecular Biology* (fall, one guest lecture, 16 students)
New course: Interdisciplinary Biological Sciences: *Graduate Computational Biology Bootcamp* (fall, 3 days, eight hours per day, 16 students) - www.GitHub.com/AndersenLab/IBiS-Bootcamp
Interdisciplinary Biological Sciences 423: *Ethics of peer review* (spring, one guest lecture, 42 students)

2013 Interdisciplinary Biological Sciences 402: *Eukaryotic Molecular Biology* (fall, one guest lecture, 24 students)

K-12 advising:

Esha Sharma, Richard Montgomery High School (2025-2026)
Ariel Shi, Baltimore Polytechnic Institute (2025-2026)
Cecilia Soko, Baltimore Polytechnic Institute (2025-2026)
Arya Takalkar, Thomas Jefferson High School (2025)
Colin Lily, Baltimore Polytechnic Institute (2024-2025)
Jinghan (Jonathan) Xiao, Lake Forest Academy (2022-2023)
Anwyn Zhou, New Trier High School (2022)
Richelle Lee, Adlai E. Stevensen High School (2022)
Preeti Rao, Adlai E. Stevensen High School (2022)
Ned Koh, Lake Forest Academy (2021-2022)
Yahya Junejo, Hinsdale Central High School (2020)
Shanthi Hegde, Lambert High School (2020 - 2021)

Aarnav Patel, Barrington High School (2020)
Justine Rozenich, Saint Ignatius College Preparatory (2020-2022)
Hannah Ahmed, Normal Community High School (2019-2020)
Ally Bardas, New Trier High School (2019-2021)
Regeneron Science Talent Search Finalist 2021
Sarosh Nagar, Glenbrook North High School (2018-2019 academic year)
Britney Sun, Glenbrook North High School (summer 2018)
Ethan Schonfeld, Glenbrook North High School (summer 2018)
Lillian Tushman, Oak Park and River Forest High School (2016-2017 academic year)
Recipient of 3rd place All-Illinois Science Research Competition
Caitlin Westerfield, Evanston Township High School (2015-2016 academic year)
Matteo di Bernardo, Evanston Township High School (2015-2016 academic year)
Recipient Columbia University Scientific Scholars Fellowship, Recipient Fulbright Fellowship
Ainsley Tran, Oak Park and River Forest High School (2015-2016 academic year)
Lauren Mann, Oak Park and River Forest High School (2014-2015 academic year)
Jacob Cruger, Latin School of Chicago (summers 2013, 2014)
Gina Liu, Illinois Math and Science Academy (2013-2014 academic year)

Undergraduate advising:
Sarah Ewig (2026 - , Class of 2029), Cell and Molecular Biology Major
Marie-Jose Jean Rohayem (2025 - , Class of 2028), Cell and Molecular Biology Major
Nora Ewis (2026, Class of 2027), Emory University
Leah Froemming (2026 - , Class of 2027), Cell and Molecular Biology Major
Olivia Jonathan (2026, Class of 2027), Boston University
Jack Weinstein (2025 - 2026, Class of 2027), Haverford College
Grayson Benson (2025 - 2026, Class of 2027), Oberlin College
Kate Stone (2024 - , Class of 2027), Cell and Molecular Biology Major
2025 Johns Hopkins Summer Provost's Undergraduate Research Award (PURA) recipient
Rohan Shrishrimal (2024 - , Class of 2028), Cell and Molecular Biology Major
Rachel Choo (2023 - 2026, Class of 2026), Cell and Molecular Biology Major
Joyce Reyes-Otero (2024, Class of 2025), Industrial Microbiology Major, Univ. of Puerto Rico - Mayagüez
2024 Johns Hopkins Biology Summer REU recipient
Alyssa Blanco (2023 - 2024, Class of 2025), Biological Sciences Major, Northwestern University
2023 PBS Summer grant recipient
2024 Johns Hopkins Biology Summer REU recipient
Fiona Shao (2022 - 2023, Class of 2024), Biological Sciences Major
2023 Summer URG recipient
Gracie Paredas (2022 - 2023, Class of 2025), Biological Sciences Major
2023 Summer URG recipient
Michael Chen (2022, Class of 2024), Biological Sciences Major, University of Southern California
Crystal Lovato (2022, Class of 2023), Biological Sciences Major, Idaho State University
2022 SROP student
Andrew Yang (2022, Class of 2025), Math Major, Brown University
2022 Quantitative Biology REU student
Sharik Khan (2022 - 2023, Class of 2024), Biological Sciences Major
2022 Summer URG recipient, 2023 Summer URG recipient
Skyler Stone (2022 - 2024, Class of 2025), Biological Sciences Major
2022 Posner Fellow, 2023 Summer URG recipient
Andrea Phung (2021 - 2024, Class of 2025), Biological Sciences Major
2022 AYURG recipient, 2022 Summer URG recipient

Cassia Yeo (2021 - 2024, Class of 2025), Biological Sciences Major
2022 Summer URG recipient, 2022 Academic Year URG recipient,
2023 Summer Advanced URG recipient, 2023 National Collegiate Undergraduate Research Award
Chloe Sokol (2021 - 2022, Class of 2024), Data Sciences Major
Nicole Banks (2021 - 2022, Class of 2022), Biological Sciences Major
Joey Gallindo (2021, Class of 2024), Engineering Major
2021 Summer URG recipient
Jordan Vaughn (2020 - 2021, Class of 2023), Biological Sciences Major
2021 WCAS Baker Family Summer Grant recipient
Karan Gowda (2020 - 2021, Class of 2022), Biological Sciences Major
2020 Center for Quantitative Biology Summer Grant recipient, 2021 Academic Year URG recipient
Iris Swarthout (2020 - 2021, Class of 2023), Biological Sciences Major
2020 Summer Internship Grant Program recipient, 2021 Summer URG recipient
Raghav Gupta (2020, Class of 2022), Biological Sciences Major
Isabella Miller (2020 - , Class of 2023), Biological Sciences and Hispanic studies double major, Bowdoin Coll.
2022 Summer research grant recipient
Emily Jahn (2019 - 2020, Class of 2020), Biological Sciences Major
2019 Weinberg College Summer Grant recipient
Kailyn Parham (2019 - 2021, Class of 2021), Biological Sciences Major
2019 Summer URG recipient, 2020 Academic Year URG Recipient, 2021 Acad. Year URG Recipient
Anna Derrick (2019 - 2020, Class of 2021), Biological Sciences Major
2019 Weinberg College Summer Grant recipient
Jake Mostad (2018 - 2020, Class of 2020), Biological Sciences Major
2019 Summer URG recipient
Katie Introcaso (2018 - 2022, Class of 2022), Biological Sciences Major
2020 Summer URG recipient, *2021 WCAS Baker Family Summer Grant recipient*
Ellen Chao (2018 - 2021, Class of 2021), Biological Sciences Major
2018 Summer URG recipient, 2019 Chemistry of Life Processes Lambert Fellow
Karol Bisaga (2017 - 2019, Class of 2020), Biological Sciences Major
2017 NU Bioscientist Summer Grant recipient
Tim Sheng (2018 - 2019, Class of 2019), Biological Sciences Major
2018 Summer URG recipient
Grace Park (2016 - 2019, Class of 2019), Biological Sciences Major
2016 Posner Fellowship recipient, 2017 Program in Biological Sciences Summer Grant recipient,
Natural Sciences and Engineering 2nd place poster prize recipient
Zyneb Adewusi (2018, Class of 2018), Biological Sciences Major, Moraine Valley Community College
2018 Summer Research Opportunities Program (SROP) recipient 2019 Northwestern Undergraduate
Yihong Hu (2018 - 2019, Class of 2021), Biological Sciences Major
2018 Program in Biological Sciences Summer Grant recipient
Chido Chikuturudzi (2017-2018, Class of 2018), Biological Sciences Major, Northeastern Illinois Univ.
Peter Finnegan (2017 - 2018, Class of 2020), Biological Sciences Major
2017 Program in Biological Sciences Summer Grant recipient
Kimberly Collins (2016 - 2018, Class of 2020), Biological Sciences Major
2017 NU Bioscientist Summer Grant recipient
Selina Deiparine (2016 - 2018, Class of 2018), Biological Sciences Major
2016 Summer URG recipient, 2016 Academic URG recipient
Rohit Rastogi (2016 - 2017, Class of 2019), Computer Science and Statistics Majors
Sarah Bier (2016 - 2017, Class of 2019), Biological Sciences Major
2016 Summer URG recipient
Mattlyn Cordova (2016 - 2017, Class of 2019), Gender Studies Major
2016 Program in Biological Sciences Summer Grant recipient
Joshua Roberts (2015-2016, Class of 2016), Computer Science Major

Nicholas Irons (2015, Class of 2018), Physics Major
2015 Summer URG recipient
Annika Zhang (2014-2015, Class of 2018), Biological Sciences Major
2015 Weinberg College Summer Grant recipient
Tyler Shimko (summers 2012, 2013, 2014, 2015, University of Utah Class of 2015), Biology Major
Barry Goldwater Scholarship recipient
Myriad Academic Scholarship recipient
Thomas Verender Hanks Scholarship recipient
National Science Foundation Graduate Research Fellowship recipient
Department of Energy Computational Science Graduate Fellowship Honorable Mention
Mazeed Aro-Lambo (2014, Class of 2017), Biological Sciences Major
2014 NU Bioscientist Summer Grant recipient
Stevie Hippleheuser (2014 - , Class of 2017), Biological Sciences Major
2016 Program in Biological Sciences Summer Grant recipient
2015 Summer URG recipient
2014 Weinberg College Summer Grant recipient
Camille Calvin (2014, Class of 2017), Mechanical Engineering Major
2014 Posner fellowship recipient
Hillary Tsang (2013 - 2016, Class of 2016), Biological Sciences Major
2015 Weinberg Summer Grant recipient
2014 Summer URG recipient
2014 Academic URG recipient
Lautaro Clienti (2013 - 2015, Class of 2017), Mechanical Engineering Major
2014 Academic URG recipient
Kreena Patel (2013 - 2015, Class of 2015), Biological Sciences and Psychology Double Major
2015 Emmanuel Margoliash Prize for Basic Research recipient
Winfred Hill Award recipient
James Alton James Scholar
Ellen Taus Scholarship recipient
J.G. Nolan Scholarship recipient
2014 Academic URG recipient
Zifan Xiang (2014 - 2015, Class of 2015), Biomedical Engineering Major
Stephen Chan (2013 - 2014, Class of 2014), Computer Science Major
2013 Summer URG recipient

Graduate student and post-doctoral advising:

Masters student advising:
Ayeh Khorshidian (Johns Hopkins Genetics and Genomics Masters), 2025
Caroline Bond (Quantitative and Systems Biology Masters), 2021-2022
Anita Huang (Biotechnology Program), 2018 - 2020
Ryan (Heechul) Chung (Quantitative and Systems Biology Masters), 2018 - 2019
Suma Aldakeel (advisor, Cindy Voisine - Northeastern Illinois Univ.), 2016 - 2017
Kristen Larrichia (advisor, Nyree Zerega – Program in Plant Biology and Conservation), 2014 - 2015
Lucie Bastin-Heline (Master's exchange student, Ecole Normale Superior, Paris, France), 2014

Graduate PhD candidates *(Current position)*:
Lina Blanco (2026 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program
Joy Mompi (2026 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program
Rose Runyan (2026 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program

Etta Schaye (2025 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program
Recipient of Victor Corces Best TA award - Development Genetics lab (2025)
Recipient of an Exploration and Field Research Grant for New Zealand field work from The Explorers Club Washington

Maya Mastranardo (2024 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program
Recipient of CMDB Margolis Travel award (2025)

Lance O'Connor (2024 -), Ph.D. student, Cell, Molecular, Biophysics, and Developmental Biology Program
NSF GRFP Honorable mention
Recipient of Victor Corces Best TA award - Biochemistry lab (2024)

Ryan McKeown (2022 -), Ph.D. student, Interdisciplinary Biological Sciences Program
Funded by the NIH Biotechnology Training grant full member (2022-2024)

Nic Moya (2020 - 2025), Ph.D. student, Interdisciplinary Biological Sciences Program
Funded by the NIH Biotechnology Training grant full member (2021-2022)

Joy Nyaanga (2019 - 2022), Ph.D. student jointly advised with Niall Mangan, Interdisciplinary Bio. Sci. Program
Funded by NSF-Simons Center for Quantitative Biology (2019-2022)
(Senior scientist, Evozyne)

Loraina Stinson (2019 - 2021), Ph.D. student - left with Masters, Interdisciplinary Biological Sciences Program
Funded by the Cell and Molecular Basis of Disease NIH Training grant (2019-2021)

Clayton Dilks (2018 - 2021), Ph.D. student, Interdisciplinary Biological Sciences Program
Funded by the NIH Biotechnology Training grant cluster (2019)
Funded by the NIH Biotechnology Training grant full member (2020)
Recipient of travel award from WAAVP (2019)
Recipient of Burroughs-Wellcome Travel Award (2019)
(Senior scientist, Evozyne)

Ye Wang (2017 - 2019), Visiting Ph.D. student, Sichuan Agricultural University, China
Funded by China Scholarship Council (2017-2019)
(Program Director for Panda Genomes, the Chengdu Panda Base)

Kathryn Evans (2016 - 2020), Ph.D. student, Interdisciplinary Biological Sciences Program
Funded by the NSF-Simons Center for Quantitative Biology (2020)
Funded by the Cell and Molecular Basis of Disease NIH Training grant (2017-2019)
Recipient of travel awards from IBiS and the Northwestern Graduate School (2017)
Recipient of Biotechnology NIH Training grant cluster member, declined (2017)
(Bioinformaticist, Precision Biosciences, Inc.)

Shannon Brady (2015 - 2019), Ph.D. student, Interdisciplinary Biological Sciences Program
Recipient of the Dr. John N. Nicholson Fellowship (2018-2019)
Funded by the Biotechnology NIH Training grant (2015-2017)
National Science Foundation Graduate Research Fellowship Program (Honorable Mention)
Recipient of travel award from Union Biometrica (2016), IBiS and the NU Graduate School (2017)
Poster first prize winner Northwestern Computational Research Day (2017)
Poster prize winner 21st International C. elegans meeting
Best TA award IBiS Graduate Program (2017)
(Consultant, Boston Consulting Group)

Daniel Cook (2014 - 2018), Ph.D. student, Driskill Graduate Program
Current position: Senior Programmer at Google, Mountain View, CA
Northwestern Graduate School Outstanding Thesis Award (2018)
Funded by a National Science Foundation Pre-doctoral Fellowship (2015-2018)
Northwestern Presidential Fellowship Finalist (2017)
Recipient of travel awards from IBiS and the Northwestern Graduate School (2016)
(Genomics team lead, Google)

Stefan Zdraljevic (2014 - 2019), Ph.D. student, Interdisciplinary Biological Sciences Program
Recipient of the Widom Award for Research Excellence (2019)
Northwestern Graduate School Outstanding Thesis Award (2019)
Recipient of the IBiS Rappaport Award for Research Excellence (2018)
Funded by the Cell and Molecular Basis of Disease NIH Training grant (2015-2017)
Recipient of travel awards from Northwestern Center for Genetic Medicine (2016, 2017)
Recipient of travel awards from IBiS and the Northwestern Graduate School (2015, 2017)
Chemistry of Life Processes Drug Discovery Scholar (2017)
(NIH NRSA F32 post-doctoral fellow at UCLA and HHMI, Kruglyak lab)

Additional rotation graduate students:
Rose Runyan (2026), Lina Blanco (2026), Zhiyu Zheng (2026), Jailyn Loor (2025), Jessica Shaffer (2025), Joy Mompi (2025), Etta Schaye (2025), Joshua Bauman (2025), Sara Seegers (2025), Izabella Mastroianni (2024), Mingling (Derek) Hui (2024), Tabor Rodriguez (2024), Francesca Batelli (2024), Jake Gavin (2024), Maya Mastronardo (2024), Lance O'Connor (2023), Corinne Croslyn (2022), Ryan McKeown (2022), Brooke Angel (2021), Jack Sumner (2021), Nic Moya (2020), Joy Nyaanga (2019), Loraina Stinson (2019), Emily Czajkowski (2018), Julie Liang (2018), Elan Ness-Cohn (2018) Kyle Siegel (2018), Clayton Dilks (2018), Garth Fisher (2017), Nicholas Sepulveda (2017), Michael Schamber (2017), Evan Buechel (2016), Nic Daffern (2016), Bryan Eder (2016), Ryan Abdella (2015), Erin Baker (2014), Alex Karge (2014), Saiorse McSharry (2014), Amy Nilles (2013), Ian Wolff (2013)

Post-doctoral researchers *(Current position)*:
Connor Leong (2026 -), Ph.D. from Oregon State Univ., advisors Drs. Thomas Sharpton and Michael Kent
Zihao (John) Li (2026 -), Ph.D. from the University of Pennsylvania, advisor Dr. Chris Fang-Yen
Nikita Jhaveri (2024 - 2026), Ph.D. from McMaster University, advisor Dr. Bhagwati Gupta
Moore Foundation Post-doctoral grant
(Visiting Teaching Track Assistant Professor, Arkansas State University)
Bowen Wang (2024 - 2025), Ph.D. from Northeast Forestry University, advisor Dr. Ling Ma
Amanda Shaver (2022 -), Ph.D. from Univ. of Georgia, advisor Dr. Arthur Edison
Janelia Leading Edge Fellow, 2024
Chair of the National Post-doctoral Council
NIH F32 Post-doctoral fellow (2024-2027)
NIH K99 Pathway to Independence Awardee (2026-2031)
José Luis Tellez Arreola (2022), Ph.D. from Universidad Autonomous de Mexico, advisor Dr. Ataulfo Torres
(Research Assistant Professor, National University Autonomous of México)
JB Collins (2021 - 2026), Ph.D. from Univ. of Georgia, advisor Dr. Ray Kaplan
USDA Post-doctoral fellow (2024-2027)
(Assistant Professor of Veterinary Clinical and Life Sciences, Utah State University)
Janneke Wit (2019 - 2021), Ph.D. from Aarhus Univ., advisor Dr. Volker Loeschcke
(Research Fellow, Aalborg University)
Sam Widmayer (2019 - 2022), Ph.D. from North Carolina State Univ., advisor Dr. David Aylor
(Associate Computational Scientist, The Jackson Laboratory)
Gaotian Zhang (2018 - 2023), Ph.D. from Ecole Normale Superieure, Paris, advisor Dr. Marie-Anne Felix
(La Fondation pour la Recherche Médicale, CNRS, Ecole Normale Superior - Paris)
Timothy Crombie (2017 - 2023), Ph.D. from University of Florida, advisor Dr. David Julian
(Assistant Professor of Biology, Florida Institute of Technology)
Lewis Stevens (2019 - 2020), Ph.D. from Univ. of Edinburgh, advisor Dr. Mark Blaxter
(Bioinformaticist, Wellcome Trust Sanger Institute, Hinxton, England)

Steffen Hahnel (2017 - 2018), Ph.D. from Justus-Liebig University, advisor Dr. Christoph Grevelding
Recipient of DFG German Science Fellowship (2018-2020)
Recipient of Northwestern Post-doctoral Travel Award (2018)
Recipient of Burroughs-Wellcome Travel Award (2018)
(Scientist, Boehringer Ingelheim Health, Germany)
Daehan Lee (2017 - 2020), Ph.D. from Seoul National University, advisor Dr. Junho Lee
(Assistant Professor of Biological Sciences, Sungkyunkwan University, South Korea)
Mostafa Zamanian (2014 - 2016), Ph.D. from Iowa State University, advisor Dr. Timothy Day
Recipient of NIH/NIAID K22 Career Transition Award (2016)
Recipient of Northwestern Post-doctoral Travel Award (2016)
Funded by the Bill and Melinda Gates Foundation (2014)
(Associate Professor of Pathobiological Sciences, University of Wisconsin - Madison)
Bryn Gaertner (2014), Ph.D. from University of Oregon, advisor Dr. Patrick Phillips
(Associate Scientific Director at Ashfield, part of UDG Healthcare PLC)

Graduate Board Oral (GBO) Exam committee memberships:
2025 (1 Biophysics): Smriti Chibber
2025 (5 CMDB): Tabor Rodriquez, Olivia Choi, Caroline Moore, Derek Hui, Izabella Mastroianni
2025 (1 CS): Kavya Vaddadi (Chair)
2024 (5 CMDB): Cameron Allen, Jake Galvin, Betty Huang, Alexis Mottram, Qi Wang
2024 (1 BME): Mao-Jan Lin (Chair)

Graduate thesis committee memberships:
Joshua Bauman (advisor, Rajiv McCoy) (chair) 2026 -
Betty Huang (advisor, Rajiv McCoy) (chair) 2024 -
David Bass (advisor, Steven Salzberg) (chair) 2024 -
Logan Wallace (advisor, Xin Chen) chair) 2024 -
Amanda Ray (advisor, Andrew Gordus) 2025
Kavya Vaddadi (advisor, Ben Langmead) (chair) 2025
Calvin Runnels (advisor, Andrew Gordus) 2024 -
Harjit Khaira (advisor, Nichole Broderick) (chair) 2024 -
Mao-Jan Lin (advisor, Ben Langmead) (chair) 2024
Nick Wong (advisor, Will Ludington) 2024 -
Sara Carioscia (advisor, Rajiv McCoy) (chair) 2024 - 2025
Corinne Croslyn (advisor, Shelby Blythe) 2022 - 2023 (no longer IBiS preceptor)
Feihong Xu (advisor, Luis Amaral) (chair) 2021 - 2023 (no longer IBiS preceptor)
Elias Guan (advisor, Chris Petersen) (chair) 2021 - 2023 (no longer IBiS preceptor)
Reese Richardson (advisor, Luis Amaral) (chair) 2021 - 2023 (no longer IBiS preceptor)
Idalis Ramirez (advisor, Rick Morimoto) 2021
Taojunfeng Su (advisor, Neil Kelleher) (chair) 2021 - 2023 (no longer IBiS preceptor)
Emily Czajkowski (advisor, Sadie Wignall) 2020 - 2023 (no longer IBiS preceptor)
Elan Ness-Cohn (advisor, Rosemary Braun - DGP, Feinberg School of Medicine) (chair) 2019 - 2021
Gabriel Cavin (advisor, Sadie Wignall) 2019 - 2022
Hannah Horton advisor, Sadie Wignall) 2019 - 2023
Alex McFarland (advisor, Erica Hartman) (chair) 2018 - 2021
Matt Robey (advisor, Neil Kelleher) 2016 - 2020
Rachel Bakker (advisor, Rich Carthew) 2015 - 2020
Joseph Muldoon (advisors, Neda Bagheri and Josh Leonard) (chair) 2015 - 2020
Aaron Sue (advisor, Thomas O'Halloran) 2014 - 2022
Ritika Giri (advisor, Richard Carthew) 2013 - 2020
Sumach Aldakeel (advisor, Cindy Voisine - Northeastern Illinois Univ.) 2016 - 2017
Adam Hockenberry (advisors, Luis Amaral and Michael Jewett) 2015 - 2017

Sarah Stainbrook (advisor, Keith Tyo) 2015 - 2019
Timothy Toby (advisor, Neil Kelleher) 2015 - 2018
Rose Njoroge (advisor, Sarki Abdulkadir – DGP, Feinberg School of Medicine), 2014 - 2018
Keila Torre-Santiago (advisor, Sadie Wignall) 2014 - 2017
Arianne Rodriguez (advisor, Yun Wang) 2014 (Transferred to DGP)
Lilien Voong (advisor, Alec Wang) 2013 - 2017

DEPARTMENT, COLLEGE, AND UNIVERSITY SERVICE

Johns Hopkins University Department of Biology Service:

2026	Faculty search committee (Computational Biology)
	Department Vice Chair
	Faculty chalk talks, organizer
	Holiday Party organizing committee
	Dr. Kavi Rangan Faculty Mentorship Committee
2025	Department Vice Chair
	Faculty chalk talks, organizer
	Holiday Party organizing committee
	Dr. Kavi Rangan Faculty Mentorship Committee
2024	Holiday Party organizing committee
2024	Dr. Kavi Rangan Faculty Mentorship Committee
	Faculty chalk talks, organizer
2023	Faculty search committee (Open), chair
	Faculty chalk talks, organizer

Johns Hopkins University Krieger School of Arts and Sciences Service:

2026	CMDB Graduate Retreat Organizer
	KSAS Dissertation Prize Committee
	CMDB Graduate Program Admissions Committee
2025	CMDB Executive Committee
	KSAS Dissertation Prize Committee
	CMDB Graduate Retreat Organizer
	CMDB Branding and Web Design Committee
2024	KSAS Dissertation Prize Committee
	CMDB Graduate Program Admissions Committee
	CMDB Graduate Retreat Organizer
2023	CMDB Graduate Program Admissions Committee

Johns Hopkins University Service:

2026	Review Committee for Innovation Fund for Community-Academic Partnerships
	Steering committee for JHURA and ORA merger
	Limited Submissions Evaluation Committee
	Undergraduate Research Grant (PURA) reviewer
2025	Steering committee for JHURA and ORA merger
	Woodrow Wilson Fellowship Review Committee
	Limited Submissions Evaluation Committee
	Undergraduate Research Grant (PURA) reviewer
2024	Woodrow Wilson Fellowship Review Committee
	Ad hoc Pew Scholars Review Committee

Northwestern University Department of Molecular Biosciences Service:

2022 Faculty search committee, chair
2021 Faculty search committee
2019 Faculty search committee
2017 Strategic Planning committee
2016 Faculty search committee
Strategic Planning committee
2015 Faculty search committee
2014 Program Review committee

Northwestern University Weinberg College of Arts and Sciences Service:

2022 WCAS Single-cell Genomics committee
WCAS Teaching Award committee
2019 Program in Biological Sciences curriculum committee
2016 Faculty search committee for the Neurobiology Department
2014 NUIN Post-doc Association, *Interviews and Start-up packages*

Northwestern University Service:

2022 Undergraduate Research Grant review committee
Steering committee for NUSeq Facility in Northwestern Medicine
Ad hoc Limited submissions grant review panel
2021 Undergraduate Research Grant review committee
NSF-Simons Center for Quantitative Biology Pilot grant review panel
Limited Submissions Grant review committee
IBiS Graduate Admissions committee, chair
Steering committee for NUSeq Facility in Northwestern Medicine
IBiS Graduate Student Advisor
Ad hoc reviewer for Data Science Initiative grants and fellowships
2020 Limited submissions grant review panel
IBiS Graduate Admissions committee, chair
Steering committee for NUSeq Facility in Northwestern Medicine
IBiS Graduate Student Advisor
2019 Limited submissions grant review panel
NSF-Simons Center for Quantitative Biology Pilot grant review panel
IBiS Graduate Admissions committee, chair
Steering committee for NUSeq Facility in Northwestern Medicine
Masters Program in Quantitative and Systems Biology, Curriculum committee
IBiS curriculum committee
2018 IBiS Graduate Admissions committee
Steering committee for NUSeq Facility in Northwestern Medicine
Masters Program in Quantitative and Systems Biology, Curriculum committee
Poster judge, Northwestern Undergraduate Research Symposium
Ad hoc reviewer for Chemistry of Life Processes undergraduate grants and fellowships
Ad hoc reviewer for Data Science Initiative grants and fellowships
NSF-Simons Center for Quantitative Biology Pilot grant review panel
2017 Discussion organizer, Northwestern Computational Research Day
Steering committee for NUSeq Facility in Northwestern Medicine
Ad hoc reviewer for Chemistry of Life Processes undergraduate grants and fellowships
Ad hoc reviewer for Data Science Initiative grants and fellowships

2016	IBiS Computational Bootcamp for incoming graduate students
	Lurie Cancer Center American Cancer Society IRG review panel
	Steering committee for NUSeq Facility in Northwestern Medicine
	Ad hoc reviewer for Chemistry of Life Processes undergraduate grants and fellowships
	Ad hoc reviewer for Data Science Initiative grants and fellowships
2015	IBiS Graduate Admissions committee
	IBiS Retreat committee, Co-chair
	IBiS Computational Bootcamp for incoming graduate students
2014	Creation of IBiS Computational Bootcamp for incoming graduate students
	IBiS Retreat committee, Co-chair
	Poster judge, Northwestern Undergraduate Research Symposium
	IBiS Graduate Admissions committee
2013	Poster judge, Northwestern Undergraduate Research Symposium